

# ZenVoice Open-Source Counter-Strategy Plan

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## Document Purpose

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This plan turns the FluidVoice gap analysis into an actionable roadmap for building **ZenVoice** as a fully free, open-source, privacy-first, multilingual-first dictation app for macOS. It covers brand positioning, legal/authorship cleanup, model integration, feature implementation, and a phased delivery schedule. The work will be done with **fully independent code** — no copying from FluidVoice.

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## 1. Brand Foundation

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### 1.1 Brand Positioning

| Element                             | Decision                                                                                         |
|-------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Name</b>                         | ZenVoice (keep existing name; reframe around calm, focused, local-first voice)                   |
| <b>Tagline</b>                      | "Speak freely. Stay private."                                                                    |
| <b>Positioning</b>                  | The open-source, privacy-first, multilingual-first dictation app for developers and power users. |
| <b>Differentiator vs FluidVoice</b> | 100% local by default, no telemetry, no closed runtimes, multilingual-first, fully open source.  |
| <b>Target audience</b>              | Developers, writers, multilingual professionals, privacy-conscious Mac users.                    |
| <b>Personality</b>                  | Calm, precise, respectful, transparent, developer-friendly.                                      |
| <b>Voice</b>                        | Clear, concise, no hype, no dark patterns, honest about limitations.                             |

### 1.2 Brand Promise

1. **Privacy by design:** your voice, transcripts, and models stay on your Mac unless you explicitly choose otherwise.
2. **Open by default:** source code, model provenance, and build pipeline are public and auditable.
3. **Multilingual-first:** we optimize for non-English languages and code-mixed speech from day one.
4. **Developer-friendly:** modular code, documented architecture, welcoming contributions.
5. **Forever free:** no subscriptions, no pricing tiers, no feature gates.

### 1.3 Visual Identity Direction

| Element         | Direction                                                          |
|-----------------|--------------------------------------------------------------------|
| Primary color   | Calm electric blue (#3B82F6) — action and live state               |
| Secondary color | Warm graphite neutrals — surfaces and text                         |
| Accent          | Success green (#22C55E), warning amber, danger red                 |
| Typography      | System fonts (SF Pro / Inter) for native feel                      |
| Logo            | Refined Zen mark; keep circular/wave motif; add "open" feel        |
| UI theme        | Clean devtool aesthetic (Graphite v4 redesign already in progress) |

### 1.4 Domain / Repo / Community

- Public GitHub repository: `github.com/imYashChaudhary973/ZenVoice`
  - Website: simple landing page + docs (reuse existing `ZenVoice-Website` project or rebuild)
  - Community: GitHub Discussions first; Discord later if demand grows
  - Distribution: GitHub Releases + Homebrew cask
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## 2. Legal & Authorship Plan

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### 2.1 Author Consent & License Selection

| Step | Action                                                                                                                                        | Owner             |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1    | Confirm sole author / copyright holder of all ZenVoice source code                                                                            | You               |
| 2    | Audit third-party dependencies for license compatibility                                                                                      | Legal/Engineering |
| 3    | Choose open-source license (recommendation: <b>MIT</b> or <b>Apache-2.0</b> for broad adoption; <b>GPLv3</b> only if copyleft is intentional) | You               |
| 4    | Replace <code>LICENSE</code> file and update <code>README.md</code> , <code>Package.swift</code> , headers                                    | Engineering       |
| 5    | Add <code>CONTRIBUTING.md</code> with CLA or DCO (Developer Certificate of Origin)                                                            | Engineering       |
| 6    | Add <code>CODE_OF_CONDUCT.md</code>                                                                                                           | Community         |
| 7    | Re-sign all commits if necessary; ensure clean provenance                                                                                     | Engineering       |

### 2.2 License Comparison

| License    | Pros                                                     | Cons                         | Recommendation          |
|------------|----------------------------------------------------------|------------------------------|-------------------------|
| MIT        | Very permissive, commercial-friendly, easy to understand | No patent grant, no copyleft | <b>Strong candidate</b> |
| Apache-2.0 | Patent grant, permissive, corporate-friendly             | Slightly longer license      | <b>Strong candidate</b> |

| License | Pros                               | Cons                                                                  | Recommendation          |
|---------|------------------------------------|-----------------------------------------------------------------------|-------------------------|
| GPLv3   | Strong copyleft, protects openness | Deters some commercial users, incompatible with some App Store models | Use only if intentional |

**Recommendation:** Use **Apache-2.0** for ZenVoice. It gives patent protection, is corporate-friendly, and still requires attribution. It avoids the GPLv3 "viral" concerns while keeping the project truly open.

## 2.3 Avoiding License Contamination from FluidVoice

- Do **not** copy FluidVoice source code into ZenVoice.
- Do **not** run FluidVoice code through AI to generate ZenVoice code.
- Use FluidVoice only as a **functional reference** (what features exist, what models are used, how the UI behaves).
- Document design decisions in `docs/decisions/` with original rationale.
- For model integration, read public model cards and Hugging Face documentation, not FluidVoice's wrapper code.

## 3. Model Integration Plan

ZenVoice will support the same model categories as FluidVoice, but with a multilingual-first default.

### 3.1 Target Model Catalog

| Model                     | FluidVoice Name        | ZenVoice Catalog ID    | Runtime             | Languages           | Size     | Priority  |
|---------------------------|------------------------|------------------------|---------------------|---------------------|----------|-----------|
| Parakeet TDT v3           | parakeet-tdt-v3        | parakeet-tdt-v3        | CoreML / FluidAudio | 25 European + RU/UK | ~460 MB  | P1        |
| Parakeet TDT v2           | parakeet-tdt-v2        | parakeet-tdt-v2        | CoreML / FluidAudio | English             | ~440 MB  | P2        |
| Parakeet Flash            | parakeet-realtime      | parakeet-flash         | CoreML / FluidAudio | English (streaming) | ~430 MB  | P2        |
| Nemotron 3.5 Multilingual | nemotron-3.5-offline   | nemotron-3.5-offline   | CoreML              | ~40 languages       | ~530 MB  | <b>P1</b> |
| Nemotron Speech 3.5       | nemotron-3.5-streaming | nemotron-3.5-streaming | CoreML              | ~40 languages       | ~670 MB  | <b>P1</b> |
| Cohere Transcribe         | cohere-transcribe-6bit | cohere-transcribe      | CoreML              | 14 languages        | ~1.5 GB  | P2        |
| Apple Speech (legacy)     | apple-speech           | apple-speech-legacy    | SFSpeechRecognizer  | System languages    | Built-in | P2        |

| Model                 | FluidVoice Name       | ZenVoice Catalog ID   | Runtime             | Languages                          | Size     | Priority       |
|-----------------------|-----------------------|-----------------------|---------------------|------------------------------------|----------|----------------|
| Apple Speech Analyzer | apple-speech-analyzer | apple-speech-analyzer | SF Speech Analytics | EN, ES, FR, DE, IT, JA, KO, PT, ZH | Built-in | P3 (macOS 26+) |
| Whisper Tiny          | whisper-tiny          | whisper-tiny          | whisper.cpp         | 99 languages                       | ~44 MB   | P1 (fallback)  |
| Whisper Base          | whisper-base          | whisper-base          | whisper.cpp         | 99 languages                       | ~81 MB   | P1             |
| Whisper Small         | whisper-small         | whisper-small         | whisper.cpp         | 99 languages                       | ~257 MB  | P2             |
| Whisper Medium        | whisper-medium        | whisper-medium        | whisper.cpp         | 99 languages                       | ~793 MB  | P3             |
| Whisper Large Turbo   | whisper-large-turbo   | whisper-large-turbo   | whisper.cpp         | 99 languages                       | ~845 MB  | P3             |
| Whisper Large         | whisper-large         | whisper-large         | whisper.cpp         | 99 languages                       | ~1.5 GB  | P3             |

### 3.2 Integration Architecture

ZenVoice already has a clean `TranscriptionProvider` abstraction in `ZenVoiceRuntime`. Extend it:

```
ZenVoiceRuntime/
├─ TranscriptionProvider.swift      (existing protocol)
├─ WhisperTranscriber.swift        (existing)
├─ ParakeetTranscriberEngine.swift (existing)
├─ NemotronTranscriberEngine.swift (new)
├─ CohereTranscriberEngine.swift   (new)
├─ AppleSpeechTranscriber.swift    (new)
└─ ModelDownloadManager.swift      (extend existing)
```

### 3.3 Model Download & Verification

Reuse ZenVoice's existing `VerifiedModelCatalog` pattern:

1. Each model entry records: publisher, source URL, revision/branch, SHA-256, expected size, format, languages, license.
2. Downloads are HTTPS-only from pinned Hugging Face URLs.
3. Atomic install: download to staging directory, verify checksum, swap atomically.
4. Cancellation isolated from new downloads.
5. License notice carried in `THIRD_PARTY_NOTICES.md`.

### 3.4 Per-Model Implementation Notes

**Parakeet TDT v3 / v2 / Flash**

- Reuse existing `FluidAudio` dependency (already pinned in `Package.swift`).
- Add `AsrModelVersion` mapping: v3, v2, realtime.
- Verify model download URLs from Hugging Face ( `nvidia/parakeet-tdt-1.1b-v3` , etc.).
- Record exact model files, sizes, and SHA-256.

### Nemotron Speech 3.5

- Add dependency on CoreML-compatible Nemotron artifacts.
- Reference model: `BarathwajAnandan/nemotron-3.5-asr-streaming320-int8-CoreML` and offline variant.
- Implement streaming buffer management with bounded sample windows.
- Handle compute-unit fallback (ANE → GPU → CPU).

### Cohere Transcribe

- Use `FluidAudio` branch or direct CoreML model loading.
- Reference model: Cohere ASR CoreML conversion on Hugging Face.
- Map 14 language tokens explicitly.

### Apple Speech

- Wrap `SFSpeechRecognizer` for legacy path.
- Wrap `SFSpeechAnalytics` (macOS 26+) for analyzer path.
- Requires user permission; no model download.

### Whisper

- Already integrated via `whisper.cpp` `XCFramework`.
- Add GGUF support optionally for smaller quantized downloads.
- Keep existing SHA-256 verification.

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## 4. Feature Counter-Plan

### 4.1 Features to Match / Exceed FluidVoice

| FluidVoice Feature     | ZenVoice Independent Implementation                                        | Notes                                 |
|------------------------|----------------------------------------------------------------------------|---------------------------------------|
| Real-time live preview | Add top/notch overlay option alongside ZenBar; improve streaming latency   | Use own overlay manager               |
| Notch-aware overlay    | Implement <code>NotchOverlayController</code> with dynamic notch detection | Independent design                    |
| Command Mode           | Add deterministic local command engine + optional script extensibility     | Avoid FluidVoice's LLM agent approach |
| Write/Rewrite Mode     | Add selected-text capture + local deterministic rewrite                    | Optional BYO-cloud provider later     |

| FluidVoice Feature             | ZenVoice Independent Implementation                                | Notes                                             |
|--------------------------------|--------------------------------------------------------------------|---------------------------------------------------|
| Audio history                  | Already exists (encrypted); add optional WAV export                | Extend existing storage                           |
| Today-usage stats              | Surface in ZenBar and menu bar; keep content-free                  | Use existing insights                             |
| Per-app configuration          | Already exists; add prompt/script routing                          | Extend <code>ApplicationProfilePreferences</code> |
| Auto-updates                   | Add Sparkle or AppUpdater integration                              | P3                                                |
| Homebrew cask                  | Add after first public release                                     | P3                                                |
| Custom dictionary / word boost | Already exists (correction rules); add pronunciation dictionary UI | Extend voice profile                              |
| Meeting/file transcription     | Add drag-and-drop file transcription service                       | New feature                                       |
| Multiple overlay sizes         | Configurable ZenBar heights + notch sizes                          | UI work                                           |
| Adaptive theming               | Complete v4 Graphite redesign; add light/dark toggle               | In progress                                       |

## 4.2 ZenVoice-Only Differentiators

| Feature                       | Description                                                         |
|-------------------------------|---------------------------------------------------------------------|
| Encrypted everything          | AES-GCM encrypted transcripts, correction rules, and recovery audio |
| No telemetry by default       | Opt-in analytics only; no cloud unless user adds keys               |
| Multilingual-first onboarding | Language selected before model; non-English defaults                |
| Hinglish/Indic-first          | Latin-script Hinglish, native-script output, English translation    |
| Developer extensibility       | Custom post-processing scripts, open model catalog, plugin API      |
| Meaning guard                 | Deterministic refinement that never changes meaning                 |
| Open build pipeline           | CI builds, signed releases, reproducible checksums                  |

# 5. Technical Foundation Improvements

## 5.1 Architecture Cleanups

- Convert uncommitted v4 Graphite redesign into clean, committed design system.
- Add a true `TranscriptionProvider` protocol and registry.
- Split `ZenVoiceStorage` further if needed (history vs. settings vs. model cache metadata).
- Add a model-download coordinator separate from UI.

## 5.2 CI / Release Pipeline

| Item                   | Action                                                                |
|------------------------|-----------------------------------------------------------------------|
| GitHub Actions CI      | Build, run core/storage/runtime checks, package app                   |
| Deterministic checks   | Expand to cover new providers (Nemotron, Cohere, Apple Speech)        |
| Model download tests   | Weekly scheduled real-speech gate                                     |
| Release readiness gate | Keep existing <code>check-release-readiness.sh</code>                 |
| Public signing         | Use GitHub-hosted Developer ID + notarization secrets (protected env) |
| Homebrew formula       | Create <code>homebrew-tap</code> repo                                 |

## 5.3 Documentation

- Rewrite README for open-source contributors.
- Add `docs/MODEL_CATALOG.md` for every new model.
- Add `docs/CONTRIBUTING.md` , `docs/ARCHITECTURE.md` updates.
- Add `docs/PRIVACY.md` reaffirming no telemetry.

# 6. Phased Execution Roadmap

## Phase 1 — Open Source Foundation (Weeks 1–4)

**Goal:** Make ZenVoice public, trustworthy, and contributor-ready. Ship the first open-source release with parity on core dictation.

| #   | Work Item                                                                               | Effort | Deliverable                      |
|-----|-----------------------------------------------------------------------------------------|--------|----------------------------------|
| 1.1 | Finalize license (Apache-2.0 recommended) and author consent                            | 1 day  | Updated <code>LICENSE</code>     |
| 1.2 | Audit/replace proprietary branding and closed-source decisions                          | 2 days | Clean repo, public README        |
| 1.3 | Commit or revert v4 Graphite redesign; finalize design tokens                           | 3 days | Merged design system             |
| 1.4 | Add <code>CONTRIBUTING.md</code> , <code>CODE_OF_CONDUCT.md</code> , issue/PR templates | 2 days | Community docs                   |
| 1.5 | Set up public GitHub repo and CI                                                        | 2 days | Public repo + Actions            |
| 1.6 | Expand model catalog: Parakeet TDT v3 + Whisper Base/Small                              | 1 week | Verified catalog entries         |
| 1.7 | Multilingual-first onboarding refactor                                                  | 4 days | New onboarding flow              |
| 1.8 | Release v0.3.0 open-source alpha                                                        | 2 days | GitHub Release + notarized build |

**Phase 1 total effort:** ~4 weeks (1 person full-time)

## Phase 2 — Multilingual Powerhouse (Weeks 5–10)

**Goal:** Match FluidVoice's model diversity and become the best multilingual dictation app. Add Nemotron, Cohere, Apple Speech, and real-time streaming UX.

| #   | Work Item                                              | Effort  | Deliverable                     |
|-----|--------------------------------------------------------|---------|---------------------------------|
| 2.1 | Integrate Nemotron 3.5 Offline + Streaming             | 2 weeks | Nemotron transcription provider |
| 2.2 | Integrate Parakeet TDT v2 + Flash (English fast paths) | 1 week  | Parakeet v2/Flash provider      |
| 2.3 | Integrate Cohere Transcribe                            | 1 week  | Cohere provider                 |
| 2.4 | Integrate Apple Speech legacy + analyzer fallback      | 1 week  | Apple provider                  |
| 2.5 | Real-time live preview + top overlay / notch support   | 2 weeks | New overlay system              |
| 2.6 | Hinglish/Indic quality improvements and benchmarks     | 1 week  | Benchmark report                |
| 2.7 | Audio history export + meeting/file transcription MVP  | 1 week  | File transcription view         |
| 2.8 | Release v0.4.0                                         | 3 days  | GitHub Release                  |

**Phase 2 total effort:** ~6 weeks (1 person full-time)

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## Phase 3 — Distribution & Advanced Features (Weeks 11–16)

**Goal:** Ship a polished, competitive product with distribution channels and advanced modes.

| #   | Work Item                                                              | Effort  | Deliverable                               |
|-----|------------------------------------------------------------------------|---------|-------------------------------------------|
| 3.1 | Command Mode (deterministic local commands + script plugins)           | 2 weeks | Command mode                              |
| 3.2 | Rewrite/Write Mode (selected-text rewrite)                             | 2 weeks | Rewrite mode                              |
| 3.3 | Optional local enhancement model (open Qwen/Phi) or BYO-cloud provider | 2 weeks | Enhancement settings                      |
| 3.4 | Auto-updater (Sparkle or AppUpdater)                                   | 1 week  | Auto-update                               |
| 3.5 | Homebrew cask + website landing page                                   | 1 week  | <code>brew install --cask zenvoice</code> |
| 3.6 | Custom dictionary / pronunciation dictionary UI                        | 1 week  | Dictionary view                           |
| 3.7 | Full accessibility audit, clean-device QA, notarized stable release    | 2 weeks | v1.0.0 release                            |
| 3.8 | Community: Discord/Discussions, contributor onboarding                 | ongoing | Active community                          |

**Phase 3 total effort:** ~6 weeks + ongoing community work

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## 7. Effort Summary

| Phase                | Duration         | Focus                                      | Approx. Person-Weeks |
|----------------------|------------------|--------------------------------------------|----------------------|
| Phase 1              | Weeks 1–4        | Open-source foundation, core parity        | 4                    |
| Phase 2              | Weeks 5–10       | Multilingual models, streaming UX          | 6                    |
| Phase 3              | Weeks 11–16      | Distribution, command/rewrite, stable v1.0 | 6                    |
| <b>Total to v1.0</b> | <b>~16 weeks</b> |                                            | <b>16</b>            |

With 2 engineers, this can compress to ~8–10 weeks. With part-time work, expect 6–9 months.

## 8. Risk Register & Mitigations

| Risk                                        | Impact | Mitigation                                                                |
|---------------------------------------------|--------|---------------------------------------------------------------------------|
| Relicensing from proprietary                | Medium | Confirm sole authorship; replace LICENSE; check dependency compatibility  |
| FluidAudio closed-source dependency         | Medium | Pin exact revisions; document license; prepare fallback to whisper.cpp    |
| Nemotron/Cohere model availability          | Medium | Verify Hugging Face repositories before committing; keep Whisper fallback |
| macOS version fragmentation                 | Low    | Keep macOS 14 minimum; gate newer APIs behind availability checks         |
| Contributor licensing                       | Low    | Use DCO in CONTRIBUTING.md; no CLA complexity needed                      |
| GPLv3 contamination from reading FluidVoice | High   | Treat FluidVoice as functional reference only; write independent code     |
| Build/signing complexity                    | Medium | Use existing scripts; protect secrets in GitHub Actions                   |

## 9. Success Metrics

| Metric                                 | Target                               |
|----------------------------------------|--------------------------------------|
| GitHub stars (6 months)                | 1,000+                               |
| Supported languages at v1.0            | 40+ with first-class UX              |
| Dictation latency (Nemotron streaming) | ≤ 300 ms word appearance             |
| Release cycle                          | Monthly alpha/beta, quarterly stable |
| Homebrew installs (6 months)           | 500+                                 |
| Zero telemetry complaints              | 100% opt-in or none                  |

## 10. Immediate Next Actions (This Week)

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1. **Decide license:** confirm Apache-2.0 vs MIT vs GPLv3.
  2. **Confirm sole authorship:** review all commits and third-party code in ZenVoice.
  3. **Clean uncommitted redesign:** commit or revert the v4 Graphite changes.
  4. **Make repo public** after license change.
  5. **Start Phase 1.6:** expand `VerifiedModelCatalog` with Parakeet TDT v3 + Whisper Small.
  6. **Schedule model research:** identify exact Hugging Face URLs for Nemotron, Cohere, and Parakeet v2/Flash.
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## Conclusion

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ZenVoice can become a credible open-source alternative to FluidVoice by leaning into its existing strengths (privacy, modularity, encrypted history) and aggressively closing the model-diversity and UX gaps. The plan is ambitious but feasible in ~16 weeks of focused work, starting with legal/open-source cleanup and ending with a stable v1.0 release.